

MOHAMMED OMER

Full-Stack Software Engineer | AI Engineer | Digital Systems Specialist

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SUMMARY

Full-Stack Software Engineer with hands-on experience across the **full software development lifecycle**, from requirements gathering and **system design** to **deployment and maintenance**. Delivered **17+ production-ready websites** and digital platforms using **Next.js, React, TypeScript, HTML5, CSS3, and JavaScript**, including a school website that **generated 116K+ Google Search Clicks** through technical **SEO** and performance optimization. Designed and developed a **Python-based Fee Management System**, deployed via the **Microsoft Store**, reducing administrative workload by **over 80%** for a real educational institution. Experienced in building **AI-powered applications**, including **OmerGPT**, an **LLM-integrated chatbot** platform with API integration and modern responsive interfaces. **Proficient in Python, Next.js, React, TypeScript, SQL, SQLite, MariaDB, Git, Vercel, Cloudflare, GoDaddy and Hostinger**, with expertise spanning **full-stack web development, desktop application development, database design, cloud deployment, and performance optimization**. Certified in **Prompt Engineering**, with a proven ability to identify **operational inefficiencies** and **deliver scalable, user-focused software solutions**. Fluent in English, Hindi, and Urdu.

Core Competencies: Full-Stack Development • Front-End Development • REST API Development • Desktop Application Development • AI Integration • Prompt Engineering • Process Automation • Software Architecture

EXPERIENCE

Founder & Full-Stack Developer, SphereRoot

📅 2025 - Present 📍 Remote / Hyderabad, India

- Founded and managed a digital solutions agency delivering full-stack web applications, business websites, desktop software, AI-powered solutions, and automation systems.
- Designed, developed, and deployed 17+ client projects across education, healthcare, retail, and service industries.
- Building scalable applications using React, Next.js, Node.js, TypeScript, Python, Electron, SQLite, MariaDB, and Tailwind CSS.
- Managed complete deployment lifecycle including domains, subdomains, DNS configuration, SSL certificates, Cloudflare CDN, Vercel hosting, CI/CD, and production monitoring.
- Led client requirement analysis, UI/UX planning, SEO optimization, API integrations, performance tuning, and ongoing maintenance.
- Integrated AI-powered chatbots, workflow automation, and custom business solutions, reducing manual administrative effort and improving operational efficiency.

EDUCATION

Bachelor of Computer Applications (B.C.A.)

St. Paul's Degree & PG College

📅 2022 - 2025

Intermediate - C.E.C. (Civics, Economics, Commerce)

St. Joseph Degree & PG College

📅 2020 - 2022

SKILLS

Front-End Development: HTML5, CSS3, JavaScript (ES6+), TypeScript, React, Next.js, Tailwind CSS, Responsive Design, Cross-Browser Compatibility, Web Performance Optimization.

Back-End Development: Node.js, Express.js, REST APIs, CRUD Operations, Authentication (JWT), Authorization, API Integration.

Databases: MariaDB, PostgreSQL, SQLite, SQL, Database Design, Query Optimization.

AI & Automation: Prompt Engineering, AI Integration, Chatbot Development, Workflow Automation.

Cloud & DevOps: Vercel, Cloudflare, GoDaddy, Hostinger, Git, GitHub, CI/CD Basics.

Software Engineering: SDLC, MVC Architecture, Debugging, Performance Optimization, Deployment, System Integration.

Deployment & Infrastructure: Vercel, Hostinger, GoDaddy, Cloudflare (CDN, DNS Management, SSL, Caching, Security), Repository Management & CI/CD Basics.

Design & Networking Tools: Figma (UI Wireframing, Prototyping), Cisco Packet Tracer (Network Simulation, IoT Simulation, Network Topology Design, Routing & Switching Basics).

PROJECTS

School Digital Transformation Suite | Full-Stack Web & Desktop Platform

Leading High School

- **Led the end-to-end digital transformation** of a real educational institution by designing, developing, and deploying a complete ecosystem of **web and desktop application** serving administrators.
- Built a modern, SEO-optimized school website from scratch using **Next.js and React**, featuring online admissions, contact forms, event galleries, announcements, responsive design, WhatsApp integration, Google Maps, and dynamic content management.
- Improved the school's digital presence and discoverability, generating **116K+ Google Search clicks** while significantly increasing online visibility, admissions inquiries, and community engagement.
- Designed and developed a desktop-based **Fee Management System** in Python, replacing manual fee collection, handwritten receipts, and paper-based record keeping for day-to-day school operations.
- Engineered administrative tools with **Tkinter/CustomTkinter, SQLite, ReportLab, OpenPyXL, and Pandas** to automate fee calculations, payment tracking, PDF receipt generation, Excel reporting, receipt numbering, and class-wise financial reports.
- Reduced administrative workload by over **80%** through workflow automation, improving operational efficiency and minimizing manual errors.
- Packaged and deployed the desktop application using **PyInstaller and MSIX**, enabling one-click installation, simplified updates, and reliable deployment across school computers.

Tech Stack: Next.js, React, TypeScript, Framer Motion, Figma, HTML5, CSS3, JavaScript, Python, Tkinter, CustomTkinter, SQLite, SQL, ReportLab, OpenPyXL, Pandas, PyInstaller, MSIX, Git, GitHub, Vercel, Cloudflare, GoDaddy, Technical SEO

OmerGPT – AI Chat Platform | Full-Stack Web Application

Final Year College Project (Live Production Application)

- Designed and developed a production-ready AI chatbot platform integrating Meta Llama 3.1 405B through a scalable API architecture, delivering real-time conversational AI with low-latency responses.
- Built a modern, responsive interface using Next.js, React, Tailwind CSS, and TypeScript with chat history, markdown rendering, syntax highlighting, model selection, and mobile-first design.
- Engineered secure backend services featuring API routing, streaming responses, prompt management, rate limiting, environment-based configuration, and REST API integration.
- Deployed the application to a production cloud environment with automated CI/CD, custom domain configuration, SSL, DNS management, performance optimization, and monitoring.

Live Production: chat.omer4k.com

Tech Stack: Next.js, React, TypeScript, Tailwind CSS, Node.js, Express.js, REST APIs, Meta Llama API, Vercel, Cloudflare, Git, GitHub, CI/CD, DNS Management, SSL, Custom Domain Configuration.

MAK Dental Clinic | Business Website & Digital Presence

Client Project

- Designed and developed a modern, responsive website to strengthen the clinic's digital identity, improve patient accessibility, and enhance trust through a professional online presence.
- Implemented SEO best practices, performance optimization, responsive design, Google Maps integration, contact forms, WhatsApp integration, and appointment call-to-actions.
- Created and managed branded social media content, increasing Instagram reach and growing the clinic's audience to 700+ followers while improving engagement.
- Improved overall digital presence, website impressions, and brand visibility by approximately 90% through website optimization and consistent branding.

Tech Stack: Next.js, React, TypeScript, Tailwind CSS, Node.js, Express.js, REST APIs, Meta Llama API, Vercel, Cloudflare, Git, GitHub, CI/CD, DNS Management, SSL, Custom Domain Configuration.

Certifications

IBM Data Analyst Professional Certificate — IBM

Responsive Web Design — freeCodeCamp

Professional Cloud Security Engineer (Preparation) — Google Cloud

Python for Data Science, AI & Development — DeepLearning.AI

SQL (Advanced) — HackerRank

Deep Learning with TensorFlow — Cognitive Class AI

Information Systems Auditing, Controls and Assurance — HKUST

Achievements

- Built and deployed 20+ production websites.
- Delivered projects across education, healthcare and business.
- Increased client search visibility by over 900%.

Technical Writing

- How to Make Your Boring macOS Terminal Look So Much Better — Medium (2023)